

C# PROGRAMMING LANGUAGE EXAMINATION

Multiple Choice Recall-Type Assessment | Total Items: 60

Instructions: Read each question carefully and select the best answer among the choices (A-D). Write your answers clearly on a separate answer sheet or record them as directed.

- 1. Which component of the .NET Framework manages code execution and provides core services like memory management?**
 - A. Class Library
 - B. Common Language Runtime (CLR)
 - C. Application Framework
 - D. Visual Studio IDE
- 2. Which character is allowed as the starting character of a C# identifier besides letters?**
 - A. Hyphen (-)
 - B. Dollar sign (\$)
 - C. Underscore (_)
 - D. Ampersand (&)
- 3. Which method prints text to the console and automatically moves the cursor to the next line?**
 - A. Console.Write()
 - B. Console.Read()
 - C. Console.WriteLine()
 - D. Console.OutputLine()
- 4. Which conversion method in .NET converts a string input specifically into a 32-bit integer?**
 - A. Convert.ToInt16
 - B. Convert.ToInt32
 - C. Convert.ToInt64
 - D. Convert.ToDouble
- 5. What requirement must be met when declaring an implicitly-typed variable using the `var` keyword?**
 - A. It must be declared as static.
 - B. It must be a global variable.
 - C. It must be initialized with a value upon declaration.
 - D. It must be placed inside a struct.
- 6. Which modifier is used to declare a variable whose value cannot be changed after initial assignment?**
 - A. static
 - B. readonly
 - C. const
 - D. fixed
- 7. What is the operator priority (precedence rank) assigned to addition (+) and subtraction (-)?**
 - A. Priority 1
 - B. Priority 2
 - C. Priority 3
 - D. Priority 4
- 8. Which group of operators holds the highest precedence rank among arithmetic operators?**
 - A. Addition (+), Subtraction (-)
 - B. Multiplication (*), Division (/), Modulus (%)
 - C. Assignment (=, +=)
 - D. Relational operators (>, <)

9. What is the result of the expression `14 % 4` in C#?

- A. 3
- B. 2
- C. 3.5
- D. 0

10. Which relational operator checks if two values are equal?

- A. =
- B. ==
- C. !=
- D. <=

11. What happens if a `break` statement is omitted from a matching case block inside a `switch` statement?

- A. A runtime exception is thrown immediately.
- B. Execution falls through to all subsequent case statements.
- C. The program terminates automatically.
- D. The program loops infinitely.

12. Which optional case inside a `switch` statement is executed when no other case label matches?

- A. else
- B. final
- C. default
- D. break

13. How many times is a `do-while` loop guaranteed to execute its code block?

- A. Zero times
- B. At least one time
- C. Exactly two times
- D. Indefinitely until user cancels

14. What is the fundamental difference between `break` and `continue` statements in loop bodies?

- A. `break` pauses the loop; `continue` restarts the program.
- B. `break` terminates the loop entirely; `continue` skips only the current iteration.
- C. `continue` exits the loop; `break` repeats the iteration.
- D. Both statements perform identical operations.

15. Which logical operator returns `true` if and only if ALL operands evaluate to `true`?

- A. ||
- B. &&
- C. !
- D. ^

16. In the conditional expression `expr1 ? expr2 : expr3`, which expression is evaluated if `expr1` is `false`?

- A. `expr1`
- B. `expr2`
- C. `expr3`
- D. Both `expr2` and `expr3`

17. Which advantage of using methods allows modifications to be isolated so they do not affect other program parts?

- A. Recursion
- B. Input Versatility
- C. Modifiability
- D. Overloading

18. How are named arguments passed to a method call statement?

- A. Using position index numbers
- B. Using parameter name followed by a colon `:` and the value
- C. Using ampersands `&` before each variable
- D. Enclosing values inside curly braces `{}`

19. Which method of passing arguments copies an argument's memory address into the parameter using the `ref` keyword?

- A. By value
- B. By reference
- C. By output
- D. By constant

20. What is method overloading in C#?

- A. Calling a method recursively until an exception occurs
- B. Declaring multiple methods with the same name but different parameters
- C. Inheriting a method from a derived class into a base class
- D. Overriding a virtual method in an abstract class

21. In recursive methods, what contains the base case that prevents infinite recursive calls?

- A. Recursive step
- B. Exit condition
- C. Constructor method
- D. Destructor block

22. Where are value types stored in memory during runtime execution?

- A. Heap
- B. Stack
- C. Disk storage
- D. External cache

23. When a reference type object is instantiated, what is stored directly on the stack?

- A. The complete object contents
- B. The heap memory address of the object
- C. The source code definition
- D. The compiler instructions

24. Which access modifier restricts member access solely to its containing class and derived classes?

- A. public
- B. private
- C. protected
- D. internal

25. Which characteristic is true regarding constructor methods in C#?

- A. They have a return type of void.
- B. They have exactly the same name as their parent class.
- C. They are declared using the `destruct` keyword.
- D. They cannot accept parameters.

26. What are the two special accessors that can be defined within property declarations?

- A. read and write
- B. get and set
- C. push and pop
- D. load and store

27. Which indexing rule do arrays follow in C#?

- A. One-indexing (starts at 1)
- B. Zero-indexing (starts at 0)
- C. Dynamic indexing (starts at user choice)
- D. Negative indexing (starts at -1)

28. Which loop statement provides a concise way to iterate through all items in an array without using an explicit counter variable?

- A. while
- B. do-while
- C. foreach
- D. for-in

29. Which array property returns the total number of elements contained in an array?

- A. Rank
- B. Length
- C. Size
- D. Count

30. Which String property-method inserts a value into a string starting at a specified index position?

- A. Insert(index, value)
- B. Concat(index, value)
- C. Append(index, value)
- D. Substring(index, value)

31. Which String property-method returns a Boolean value `true` if a string contains a specified target value?

- A. IndexOf(value)
- B. Contains(value)
- C. HasValue(value)
- D. Matches(value)

32. What term is used for a parent class whose members and features are inherited by a derived class?

- A. Superclass
- B. Base class
- C. Abstract interface
- D. Concrete class

33. Which access modifier prevents a class from being inherited by any other class?

- A. static
- B. sealed
- C. abstract
- D. private

34. What symbol prefixes the class identifier when declaring a destructor method in C#?

- A. Exclamation mark (!)
- B. Tilde (~)
- C. Asterisk (*)
- D. Hash (#)

35. Which keyword is used in a base class method declaration to indicate that it can be overridden in derived classes?

- A. override
- B. virtual
- C. abstract
- D. new

36. What is a primary feature of an abstract class in C#?

- A. It can be directly instantiated with the `new` keyword.
- B. It cannot be instantiated and may contain abstract methods without bodies.
- C. All of its methods must be sealed.
- D. It cannot serve as a base class.

37. Which members CANNOT be contained inside an interface declaration?

- A. Properties
- B. Methods
- C. Fields (variables)
- D. Indexers

38. What default access modifier applies to all members declared inside an interface?

- A. `private`
- B. `protected`
- C. `public`
- D. `internal`

39. What is a nested class?

- A. A class derived from an abstract interface
- B. A class declared as a member inside another class
- C. A static class that cannot contain static members
- D. A class containing only destructors

40. Which pair of keywords is used to define and consume custom namespaces in C#?

- A. `using` and `namespace`
- B. `import` and `package`
- C. `include` and `module`
- D. `require` and `space`

41. How many destructors can be defined for a single class in C#?

- A. Zero
- B. Exactly one (1)
- C. Unlimited
- D. Up to two (2)

42. What limitation applies to static methods within a class?

- A. They cannot accept arguments.
- B. They can only access static members.
- C. They must return an integer value.
- D. They cannot be called from `Main()`.

43. When is a static constructor automatically called?

- A. Every time an object is instantiated
- B. Once, when a static member of the class is accessed
- C. When the program finishes execution
- D. Whenever a destructor is invoked

44. Which static method of the `Math` class returns a specified number raised to a specified power?

- A. `Math.Sqrt()`
- B. `Math.Pow()`
- C. `Math.Abs()`
- D. `Math.Round()`

- 45. Which method of the static `Array` class sorts array elements in ascending order?**
- A. `Array.Order()`
 - B. `Array.Sort()`
 - C. `Array.Reverse()`
 - D. `Array.Arrange()`
- 46. Which static property of the `DateTime` class returns the current date and time from the system clock?**
- A. `DateTime.Today`
 - B. `DateTime.Now`
 - C. `DateTime.Current`
 - D. `DateTime.SystemClock`
- 47. What does the `this` keyword refer to inside a class implementation?**
- A. The current instance/object of the class
 - B. The base parent class
 - C. The global namespace
 - D. The static entry point method
- 48. What is a key difference between `readonly` members and `const` members?**
- A. `readonly` members can have values assigned or changed inside a constructor.
 - B. `const` members can be declared without initialized values.
 - C. `const` members can hold evaluation values.
 - D. `readonly` members are implicitly static by definition.
- 49. Which C# feature allows objects to be indexed in the same manner as array elements using square brackets `[]`?**
- A. Indexers
 - B. Auto-properties
 - C. Structs
 - D. Enumerations
- 50. Overloaded operator methods must always be declared with which access modifier?**
- A. `private`
 - B. `static`
 - C. `virtual`
 - D. `abstract`
- 51. Which keyword is used to declare value types that encapsulate small groups of related variables?**
- A. `class`
 - B. `struct`
 - C. `enum`
 - D. `interface`
- 52. What constructor restriction applies specifically to `struct` declarations?**
- A. Structs cannot contain constructors with parameters.
 - B. Structs cannot contain default parameterless constructors.
 - C. Struct constructors must be `private`.
 - D. Struct constructors must return a value.
- 53. Which keyword is used to declare a specialized type containing a named list of integral constants?**
- A. `list`
 - B. `enum`
 - C. `struct`
 - D. `array`

54. When an enum element's index value is manually assigned, what value is automatically assigned to the subsequent unassigned enum item?

- A. Zero
- B. One increment above the previous element's value
- C. Double the previous value
- D. It results in a syntax error

55. In exception handling, which code block encloses statements that might throw or generate an exception?

- A. catch
- B. try
- C. finally
- D. throw

56. Which exception type is thrown when an invalid index is used to access an element of an array?

- A. FileNotFoundException
- B. FormatException
- C. IndexOutOfRangeException
- D. InvalidOperationException

57. What is the primary purpose of the optional `finally` block in exception handling?

- A. To execute code regardless of whether an exception was thrown or caught
- B. To catch specific runtime arithmetic exceptions
- C. To suppress all errors and prevent compilation
- D. To restart program execution from `Main()`

58. Which method of the `System.IO.File` class creates a file and writes content to it, overwriting it if it already exists?

- A. File.AppendAllText()
- B. File.WriteAllText()
- C. File.ReadAllText()
- D. File.Copy()

59. Which `File` class method returns a Boolean value indicating whether a file exists at a specified path?

- A. File.HasFile()
- B. File.Exists()
- C. File.Check()
- D. File.Open()

60. Which method of the `System.IO.File` class relocates a file from a specified location to a new target directory or location?

- A. File.Copy()
- B. File.Move()
- C. File.Transfer()
- D. File.Replace()

ANSWER KEY

Item	Ans	Item	Ans	Item	Ans	Item	Ans
1	B	16	C	31	B	46	B
2	C	17	C	32	B	47	A
3	C	18	B	33	B	48	A
4	B	19	B	34	B	49	A
5	C	20	B	35	B	50	B
6	C	21	B	36	B	51	B
7	B	22	B	37	C	52	B
8	B	23	B	38	C	53	B
9	B	24	C	39	B	54	B
10	B	25	B	40	A	55	B
11	B	26	B	41	B	56	C
12	C	27	B	42	B	57	A
13	B	28	C	43	B	58	B
14	B	29	B	44	B	59	B
15	B	30	A	45	B	60	B